Practice workbook · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Methods for finding weaknesses (penetration testing, ethical hacking, analysis tools, policy review) and protecting software (code reviews, audit trails, modular testing, secure design).

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-10

# 1. Question 1 · part e · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part e only.

**1** Computers can be networked together.

(a) Complete the table by adding a cross ☒ to match **each** statement to **one** networking term.

**(4)**

| Statement | DNS | PAN | WAN | WWW |
|---|---|---|---|---|
| A network of networks **not** limited to a single location | ☒ | ☒ | ☒ | ☒ |
| A network that connects devices within the range of a single user | ☒ | ☒ | ☒ | ☒ |
| A set of hypertext documents accessed through the HTTP protocol | ☒ | ☒ | ☒ | ☒ |
| A system that maps domain names to IP addresses | ☒ | ☒ | ☒ | ☒ |

(b) Networks use the TCP/IP protocol model.

The application layer and the network layer are two layers in the TCP/IP model.

Give the names of the **other two** layers.

**(2)**

1 .....

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2 .....

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(c) Emails can be sent between computers on a network.

Identify the protocol required to **send** an email.

**(1)**

☒ **A** HTTP

☒ **B** IMAP

☒ **C** POP3

☒ **D** SMTP

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# 1. Question 1 · part e · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part e only.

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(d) A library sets up a network using a star topology.

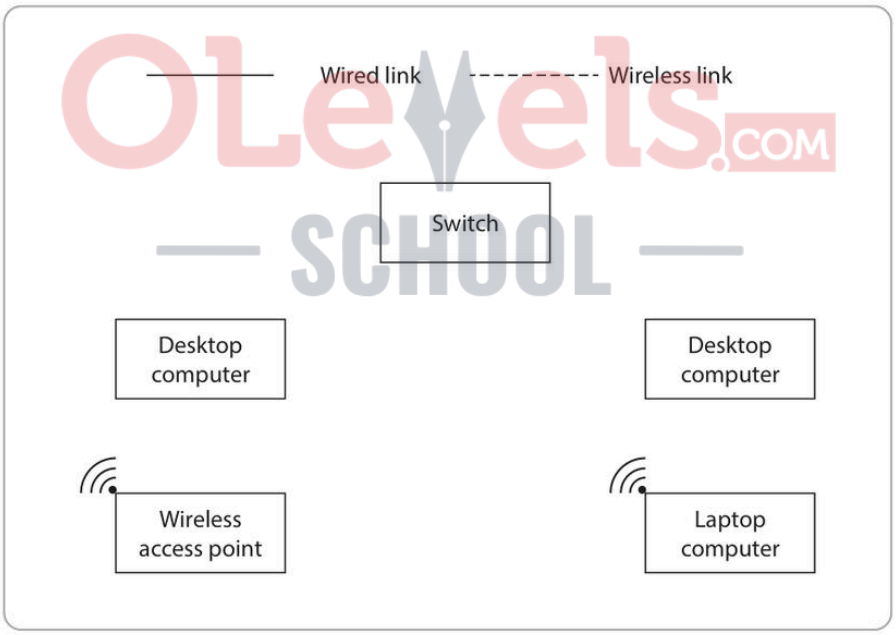
The library is connecting:

- two desktop computers with wired network interface cards
- a laptop with a wireless network interface card
- a switch
- a wireless access point.

Complete the diagram to show how the devices are connected.

Use a solid line to indicate a wired link and a dashed line to indicate a wireless link.

**(3)**




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# 1. Question 1 · part e · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part e only.

(e) IT teams carry out network penetration testing.

Describe the purpose of penetration testing.

(2)

(Total for Question 1 = 12 marks)

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Working and answer

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## 2. Question 5 · part c · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

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**5** Programmers work as part of a team to develop software.

(a) They use truth tables to determine if an expression evaluates to true or false.

Complete this truth table for the Boolean expression

**P AND (Q OR R)**

You may not need to use all rows.

**(3)**

| P | Q | R | Q OR R | P AND (Q OR R) |
|---|---|---|---|---|
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## 2. Question 5 · part c · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

(b) The team is working on software for a ride-on lawn mower.

The lawn mower must not start unless it is safe for the driver.

- The brake (B) must be engaged.
- The driver's seat sensor (S) must be engaged.
- The blade safety lock (L) must be removed.

Construct a Boolean expression, using AND, OR, and NOT with the letters B, S and L to show the safe starting condition.

**(2)**

(c) The team uses code reviews as part of the software development process.

State **two** ways in which code reviews can help protect software systems from cyber attacks.

**(2)**

1 .....

2 .....

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14

P 6 1 8 8 4 R R A 0 1 4 2 0

## 2. Question 5 · part c · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

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(d) The team needs to protect its intellectual property rights over the systems that it develops.

Give **three** types of legal protection the team could use.

(3)

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2 .....

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3 .....

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**(Total for Question 5 = 10 marks)**

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Working and answer

# 3. Question 2 · part d.i · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part d.i only.

**2** Danielle manages network security for a small business.

(a) Identify **one** action Danielle could take to protect the **network** from unauthorised access.

(1)

- ☒ **A** Give each user a separate account
- ☒ **B** Enforce a strong password policy
- ☒ **C** Make a backup plan
- ☒ **D** Encrypt sensitive files

(b) Danielle has received this email.

She is a customer of SafePayment.com but thinks this is a phishing attempt.

**From:** SafePayment.accounts@5afepayment.com
**To:** Danielle616
**Subject:** Account restricted (case SP-0011312-2021-06)

Dear Customer,

We have noticed some unusual activity on your account, so have stopped all payments.

Please use this link to log in and check your account.

Log in

Your normal log in will stoop working until you have done this.

Regards
SafePayment fraud prevention team.

(i) Danielle has circled the words **Dear Customer**.

Give **one** reason why the use of these words might make Danielle suspect that the email is a phishing attempt.

(1)

(ii) Draw circles around **two other** aspects of the email that might make Danielle suspect it is a phishing attempt.

(2)

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## 3. Question 2 · part d.i · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part d.i only.

(c) Some employees use company laptops in public places.

Danielle is concerned that shoulder surfing could pose a risk to security.

(i) Describe what is meant by **shoulder surfing**.

**(2)**

(ii) Explain **one** way to prevent shoulder surfing.

**(2)**

(d) Danielle wants to protect the network from cyberattacks.

(i) Identify the purpose of penetration testing.

**(1)**

- ☒ **A** To decompile executable code
- ☒ **B** To identify security vulnerabilities
- ☒ **C** To test the strength of an encryption algorithm
- ☒ **D** To enforce a network user policy

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## 3. Question 2 · part d.i · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part d.i only.

(ii) Danielle is sent a software patch for the network operating system but is too busy to install it straight away.

Explain why this delay could pose a threat to the security of the network.

(2)

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**(Total for Question 2 = 11 marks)**

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Working and answer

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# 4. Question 6 · part c.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.i only.

**6** Santiago manages a computer network for a small business.

(a) Networks are based on a topology.

**Figure 3** shows a network topology.

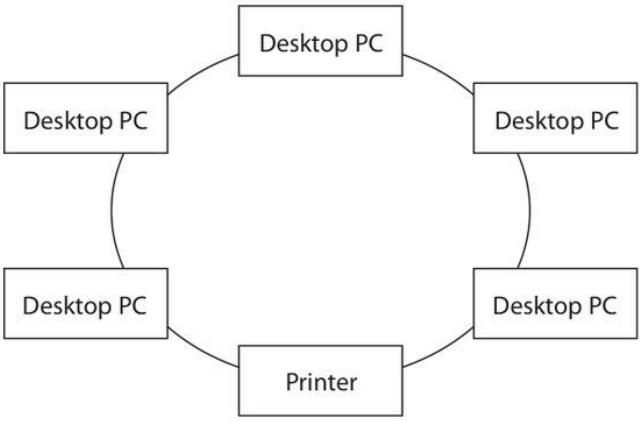



**Figure 3**

(i) Explain **one** benefit of this network topology. **(2)**

(ii) The internet is the world's largest mesh network.

Explain **one** reason why a mesh topology is essential for the internet. **(2)**

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P 7 2 5 3 8 A 0 1 8 2 0

# 4. Question 6 · part c.i · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.i only.

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(b) Santiago works on his laptop whilst travelling by train.

There is a free Wi-Fi connection on the train, but Santiago doesn't use it.

He prefers to set up a network between his smartphone and his laptop to connect to the internet.

(i) Name this type of network.

**(1)**

(ii) Explain **one** advantage for Santiago of using the network he has set up to connect to the internet, rather than the free Wi-Fi connection.

**(2)**

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(c) Santiago uses audit trails to help protect the network.

(i) State what is meant by an **audit trail**.

**(1)**

(ii) Give **one** way the data from audit trails can be used to help keep the network secure.

**(1)**

## 4. Question 6 · part c.i · 1 mark · Working and answer

Working and answer

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## 5. Question 6 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

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(a) Networks are based on a topology.

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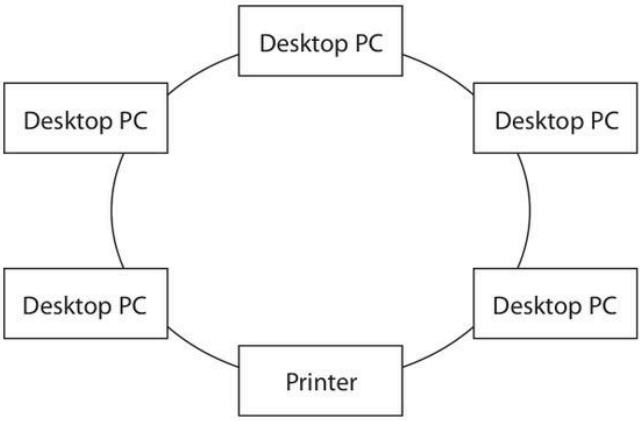



**Figure 3**

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## 5. Question 6 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

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(b) Santiago works on his laptop whilst travelling by train.

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**(1)**

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**(2)**

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(c) Santiago uses audit trails to help protect the network.

(i) State what is meant by an **audit trail**.

**(1)**

(ii) Give **one** way the data from audit trails can be used to help keep the network secure.

**(1)**

## 5. Question 6 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

(d) Santiago wants to find and fix network vulnerabilities before the reputation of the company suffers.

Discuss the methods he can use.

You should consider:

- ethical hacking
- commercial analysis tools
- review of network and user policies.

(6)

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**(Total for Question 6 = 15 marks)**

## 5. Question 6 · part c.ii · 1 mark · Working and answer

Working and answer

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# 6. Question 6 · part d · 6 marks

2022 JUN · Paper 01

Further practice · Attempt part d only.

**6** Santiago manages a computer network for a small business.

(a) Networks are based on a topology.

**Figure 3** shows a network topology.

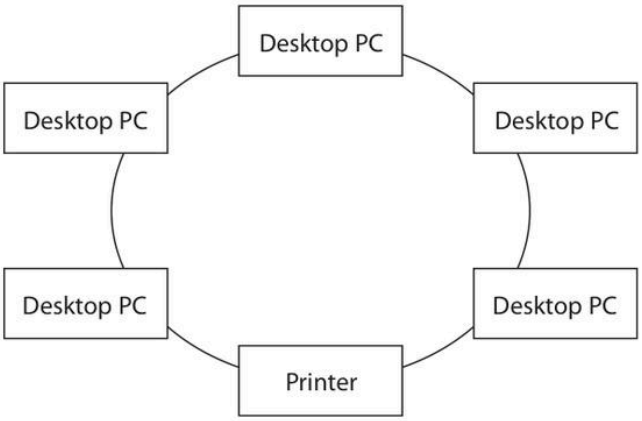



**Figure 3**

(i) Explain **one** benefit of this network topology. **(2)**

(ii) The internet is the world's largest mesh network.

Explain **one** reason why a mesh topology is essential for the internet. **(2)**

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# 6. Question 6 · part d · 6 marks

2022 JUN · Paper 01

Further practice · Attempt part d only.

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(b) Santiago works on his laptop whilst travelling by train.

There is a free Wi-Fi connection on the train, but Santiago doesn't use it.

He prefers to set up a network between his smartphone and his laptop to connect to the internet.

(i) Name this type of network.

**(1)**

(ii) Explain **one** advantage for Santiago of using the network he has set up to connect to the internet, rather than the free Wi-Fi connection.

**(2)**

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(c) Santiago uses audit trails to help protect the network.

(i) State what is meant by an **audit trail**.

**(1)**

(ii) Give **one** way the data from audit trails can be used to help keep the network secure.

**(1)**

# 6. Question 6 · part d · 6 marks

2022 JUN · Paper 01

Further practice · Attempt part d only.

(d) Santiago wants to find and fix network vulnerabilities before the reputation of the company suffers.

Discuss the methods he can use.

You should consider:

- ethical hacking
- commercial analysis tools
- review of network and user policies.

(6)

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**(Total for Question 6 = 15 marks)**

## 6. Question 6 · part d · 6 marks · Working and answer

Working and answer

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## 7. Question 1 · part d · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part d only.

**1** People use networks in all aspects of their lives.

(a) An office building has desktop computers connected in a network.

One reason for networking computers is to enable technicians to maintain them remotely.

State **two other** reasons for connecting computers in a network.

**(2)**

1 .....

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2 .....

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(b) Access to data must be controlled.

(i) Identify **one** reason that encryption is used.

**(1)**

☒ **A** To compress the file so it takes up less storage space

☒ **B** To make sure both sender and receiver use the same key

☒ **C** To make sure data is only understood by the intended receiver

☒ **D** To stop malicious hackers getting into a computer network

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## 7. Question 1 · part d · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part d only.

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(ii) File servers are in a small room at the back of an office.

The office has a burglar alarm.

Give **two other** ways that the servers can be secured using physical methods.

**(2)**

1 .....

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2 .....

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(iii) Phishing is a type of social engineering.

State what is meant by the term **social engineering**.

**(1)**

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(c) 5G is a communication standard.

Identify the characteristic that is true for the 5G communication standard.

**(1)**

- ☒ **A** It has a high transmission latency
- ☒ **B** It has built-in security
- ☒ **C** It has a lower bandwidth than 3G and 4G
- ☒ **D** It uses wireless communication

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## 7. Question 1 · part d · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part d only.

(d) Programmers write and test code.

They use modular testing.

State what is meant by the term **modular testing**.

(1)

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(e) Computer scientists are viewed as professionals, in the same way that doctors are viewed as professionals.

Give **two** ways that computer scientists can demonstrate professionalism.

(2)

1 .....

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2 .....

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**(Total for Question 1 = 10 marks)**

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Working and answer

## 8. Question 4 · part a.iii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a.iii only.

**4** A house has solar panels and a battery storage system.

(a) A smartphone application is used to monitor the battery storage system remotely. The smartphone uses a 5G connection.

(i) Give **two** benefits of a 5G connection.

**(2)**

1 .....

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2 .....

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(ii) The application requires users to authenticate their identity.

One method of authentication is a user ID and password.

Describe **one other** method the application could use.

**(2)**

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(iii) Software developers perform code reviews.

Give **two** reasons for carrying out a code review.

**(2)**

1 .....

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2 .....

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## 8. Question 4 · part a.iii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a.iii only.

(b) The battery storage system stores readings as signed 8-bit binary integers in **sign and magnitude** format.

Complete the table to show the 8-bit binary pattern and denary value for the lowest and the highest values that can be stored.

**(2)**

| | 8-bit binary pattern | Denary value |
|---|---|---|
| Lowest value | | |
| Highest value | | |

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## 8. Question 4 · part a.iii · 2 marks · Working and answer

Working and answer

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